Scientists Find Dinosaur Blood That Survived 115 Million Years

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Scientists have found traces of dinosaur blood inside bones that are 115 million years old. The discovery shows that delicate biological molecules can survive far longer than researchers once believed.

An international team led by Curtin University in Australia studied the limb bones of Aratasaurus museunacionali, a meat-eating **theropod** dinosaur found in northeastern Brazil. Inside tiny passages in the bones that once held blood vessels, the team found fossilized red blood cells. Chemical tests revealed **heme** molecules, parts of **hemoglobin**, the protein that lets red blood cells carry oxygen. The team also found amino acids linked to **collagen**, a structural protein found in bones and other connective tissues.

Earlier studies had spotted structures that looked like blood cells in ancient fossils. This is the first time scientists have found clear chemical proof of **hemoglobin**-derived molecules inside fossilized dinosaur red blood cells. The findings were published in the Proceedings of the National Academy of Sciences.

So how did the blood survive for 115 million years? Part of the answer is a mineral called **hausmannite**, which helped stabilize the fossilized cells and protect their molecules. The dinosaur was probably buried quickly in sediment with little oxygen and plenty of microbial activity, conditions that may have slowed the usual breakdown of biological material. In effect, the mineral acted like a natural time capsule.

The discovery challenges a long-held idea in paleontology that blood molecules could not survive across such vast stretches of time. Professor Kliti Grice, who led the study, called the find a landmark discovery. The team says the work opens new paths for molecular-level paleontology, letting scientists study how blood, metabolism, and even activity levels evolved in ancient animals. Other fossils may hold molecular clues that earlier examinations missed.

Word Treasure

theropod -- A theropod is a meat-eating dinosaur that walked on two legs.

heme -- Heme is a small part of hemoglobin that helps blood carry oxygen.

hemoglobin -- Hemoglobin is the protein in red blood cells that lets them carry oxygen around the body.

collagen -- Collagen is a strong protein that gives structure to bones and other tissues.

hausmannite -- Hausmannite is a mineral that helped protect the fossilized blood cells in this dinosaur.

Background you need

Dinosaurs were animals that lived on Earth long ago, and their bones can turn into fossils, which are the preserved remains of living things.

Blood is the liquid that carries oxygen and other things around an animal's body, and hemoglobin is the protein in red blood cells that lets them carry oxygen.

Paleontology is the study of ancient life through fossils, and scientists use chemical tests to learn what molecules fossils still contain.

Fossils usually form when an animal is buried quickly in mud or sand, which can protect its remains from breaking down.

Break it down

LEAD: Scientists found traces of dinosaur blood in bones 115 million years old, showing that delicate biological molecules can survive far longer than researchers once believed.

KEY EVENT: An international team led by Curtin University found fossilized red blood cells with heme molecules inside the limb bones of Aratasaurus museunacionali from northeastern Brazil.

+- WHO: An international team led by Curtin University in Australia, led by Professor Kliti Grice

+- WHAT: Fossilized red blood cells, heme molecules from hemoglobin, and amino acids linked to collagen

+- WHERE: Inside tiny passages in the limb bones that once held blood vessels

EVIDENCE: Chemical tests revealed heme molecules, and this is the first clear chemical proof of hemoglobin-derived molecules inside fossilized dinosaur red blood cells, while earlier studies had only spotted structures that looked like blood cells.

PRESERVATION: The mineral hausmannite helped stabilize the cells, and the dinosaur was probably buried quickly in sediment with little oxygen and plenty of microbial activity.

OPEN QUESTION: What other fossils may hold molecular clues that earlier examinations missed, and how blood, metabolism, and activity levels evolved in ancient animals.

Why it matters

The discovery challenges the long-held idea that blood molecules could not survive across such vast stretches of time, and it opens new paths for molecular-level paleontology, letting scientists study how blood, metabolism, and activity levels evolved in ancient animals.

Test what you remember

1. What did scientists find inside the limb bones of the 115-million-year-old dinosaur?

(A) Fossilized skin cells
(B) Fossilized red blood cells
(C) Fossilized feathers
(D) Fossilized footprints

2. Which dinosaur species did the team study?

(A) Aratasaurus museunacionali
(B) A giant long-necked dinosaur from Asia
(C) A duck-billed dinosaur from Europe
(D) A horned dinosaur from North America

3. What did chemical tests reveal inside the fossilized red blood cells?

(A) Tiny pieces of dinosaur DNA
(B) Heme molecules, parts of hemoglobin
(C) Fragments of fossilized feathers
(D) Bits of the dinosaur's last meal

4. What mineral helped stabilize the fossilized cells and protect their molecules?

(A) Hausmannite
(B) Quartz
(C) Limestone
(D) Sandstone

5. How was the dinosaur probably buried?

(A) In a river, where its bones were washed clean
(B) Slowly, in dry sand with plenty of oxygen
(C) Quickly in sediment with little oxygen and plenty of microbial activity
(D) Under a volcano, in hot ash

6. Where were the findings published?

(A) In a museum guidebook
(B) In a children's science magazine
(C) In a newspaper article
(D) In the Proceedings of the National Academy of Sciences

Answer key (try the quiz first!): 1: B 2: A 3: B 4: A 5: C 6: D

Questions to think about

1. **Professor Kliti Grice, who led the study: Called the find a landmark discovery.**

2. **The research team: Says the work opens new paths for molecular-level paleontology.**

MY THOUGHTS (at least 20 words):